Using Visual Studio Code to Connect to CERN's Ixplus Cluster

To directly edit files and run commands on CERN's Ixplus computing cluster from Windows, use Visual Studio Code paired with Microsoft's Remote - SSH extension, [**ssh˙config˙guide**, 1, 2]

1 Ensure OpenSSH is Enabled on Windows

Windows 10/11 comes with OpenSSH built-in. Verify it is active:

- Open PowerShell or Command Prompt and run: `ssh -V`
- If unrecognized, go to **Settings** → **System** → **Optional Features**, search for OpenSSH Client, and install it.

2 Install VS Code and Remote - SSH Extension

1. Install Visual Studio Code.
2. Open VS Code, press `Ctrl+Shift+X` to open the Extensions tab, search for **Remote - SSH** (published by Microsoft), and click Install.

3 Configure Your SSH Host

Ensures connection persistence through CERN load-balancers. `Ixplus.cern.ch` routes requests to dynamic, load-balanced machines. Setting up an SSH config file avoids repeated timeouts and re-authentication prompts:

1. In VS Code, press `F1` (or `Ctrl+Shift+P`) to open the Command Palette.
2. Type and select `Remote-SSH: Open SSH Configuration File...`.
3. Choose your user configuration file (usually `C:\Users\<YourWindowsUser>\.ssh\config`).
4. Add the following block (replace `your_cern_username` with your CERN computing account ID):

```
Host Ixplus
  HostName Ixplus.cern.ch
  User your_cern_username
  ServerAliveInterval 60
  ServerAliveCountMax 10
  ForwardX11 yes
  ForwardX11Trusted yes
```

5. Save and close the file.

4 Connect to Ixplus

1. Press `F1` and choose `Remote-SSH: Connect to Host...`.
2. Select `Ixplus` from the list.
3. When prompted, choose **Linux** as the target OS.
4. Enter your CERN primary account password
5. Click on `details` on the bottom left pop up to enter your 2FA/OTP if logging in from outside the CERN network.
6. VS Code will install its remote server binary in `~/.vscode-server` on your Ixplus home area.

5 Open Remote Directories and Run Code

Once connected (indicated by `SSH: Ixplus` in the bottom-left corner of VS Code):

- Go to **File** → **Open Folder...** to navigate your CERN AFS/EOS workspace (e.g., `/afs/cern.ch/user/u/username/` or `/eos/user/u/username/`).
- Open a terminal using **Terminal** → **New Terminal** (`Ctrl + ``). This shell runs directly on your assigned Ixplus node, allowing you to source environments (e.g., `asetup`, `cvmfs`, ROOT) and compile or run scripts natively.

References

[1] ATLAS Experiment. *ATLAS Software VS Code Remote SSH Guide*. Guide for setting up C++ debugging and build environments over LXPLUS SSH. 2026. URL: `https://atlas-software.docs.cern.ch/athena/ide/ssh/`.

[2] CERN Service Portal. *KB0008901: Lxplus and Visual Studio Code*. CERN-specific guidelines and Kerberos integration details. 2026. URL: `https://cern.service-now.com/service-portal?id=kb_article&sys_id=KB0008901`.